HARRISON B. GOLDBERG

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Mechanical Engineering Student | Robotics & Electromechanical Systems

EDUCATION

Arizona State University, Tempe, AZ Expected May 2029
B.S. Mechanical Engineering | GPA: 3.37/4.00

Technion – Israel Institute of Technology, Haifa, Israel 2024 – 2025
Visiting Engineering Coursework

INDEPENDENT ENGINEERING PROJECTS

Custom Robotic Actuators – Planetary & Cycloidal Gearboxes 2026 – Present
Independent Project | Actuator Design & Integration Owner

- Owned actuator builds from parametric Fusion CAD and measured-part fits through printing, assembly, electronics integration and bench testing; built 9.97:1 planetary and 25:1 cycloidal reducers.
- Wired ESP32-S3, TMC2209 STEP/DIR drive, NEMA17 motor and 14-bit MT6701 I²C output encoder; integrated 24 V motor and 3.3 V logic power.
- Specified and closely guided AI-generated C++/Arduino firmware for timer-driven stepping, encoder feedback, USB serial telemetry and fault handling; directed debugging from bench observations.
- Reduced nominal clearances to lower mechanical lost motion from 71 to 57 arcmin; separate calibrated, unloaded closed-loop tests measured 1.98 arcmin mean reversal error across four angles. Latest firmware hardware revalidation, torque and durability tests pending.

Bitzy – Five-Axis Modular Robotic Arm June 2026 – Present
Independent Project | Mechanical Design, Electronics & Firmware Integration

- Designed, printed and assembled a five-axis, six-servo arm with paired shoulder actuators, bearing supports and interchangeable tooling; built a powered rack-and-pinion gripper with a 2–30 mm design opening.
- Integrated ESP32-S3/PCA9685 PWM/I²C servo control; directed AI-generated C++/Arduino firmware for coupled shoulder commands, pulse limits, prepare/arm sequencing and heartbeat-triggered disarming.
- Directed AI-assisted Python control software for tool-specific inverse kinematics, teach/replay and calibration; diagnosed physical reference/command mismatches and drove coordinated mechanical, firmware and UI revisions.

Formivect – Scriptable CAD Automation Platform September 2026 – Present
Independent Project | Creator & Technical Direction

- Defined requirements, workflows and acceptance checks for AI-generated Fusion add-in and MCP integration exposing full Python-scripted access to the native CAD API.
- Guided parametric modeling, parameter changes, interference reviews and STEP/STL export workflows; reviewed feature health and regeneration results. Automated tolerance reasoning and independent simulation remain goals.

ADDITIONAL EXPERIENCE

TinkerTank – Founder & Product Designer 2025 – Present

- Design, prototype, manufacture and sell PETG aquarium components on Etsy; own customer requirements, parametric CAD, physical fit iterations and order fulfillment.

Engineering Design Club – The Heschel School, NYC 2022 – 2024

- Founded and co-led the club; taught Tinkercad, Fusion 360, 3D printing and additive-manufacturing workflows.

TECHNICAL SKILLS

Mechanical & CAD: Fusion 360, SolidWorks, parametric assemblies, fit/clearance iteration, FDM (PAHT-CF/PETG-CF)
Electronics & Test: Soldering, ESP32-S3, PCA9685, TMC2209, MT6701, I²C, STEP/DIR, MATLAB, encoder bench testing
Firmware & Software: C++/Arduino and Python (AI-generated implementation; closely directed and reviewed); PlatformIO, serial control, test logging